THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The largest number that divides 86 and 133 leaving remainders 2 and 7 respectively is: 1

(a) 44 **(b)** 41 **(c)** 42 **(d)** 43

2. The least number which when divided by 5, 10 and 15 leaves remainder 1 each time is: 1

(a) 29 **(b)** 30 **(c)** 32 **(d)** 31

3. The decimal expansion of $\frac{13}{75}$ is: 1

(a) non-terminating repeating **(b)** terminating

(c) an integer **(d)** non-terminating non-repeating

4. If $\sqrt{11}$ is irrational, then $\sqrt{11} \times \sqrt{11}$ is: 1

(a) rational **(b)** irrational

(c) non-terminating non-repeating **(d)** not defined

5. The HCF of 42, 63 and 105 is: 1

(a) 23 **(b)** 21 **(c)** 20 **(d)** 22

6. The LCM of 14, 21 and 28 is: 1

(a) 85 **(b)** 84 **(c)** 42 **(d)** 823

7. The largest number that divides 97 and 172 leaving remainders 7 and 7 respectively is: 1

(a) 17 **(b)** 16 **(c)** 14 **(d)** 15

8. The least number which when divided by 8, 12 and 20 leaves remainder 5 each time is: 1

(a) 115 **(b)** 125 **(c)** 120 **(d)** 126

9. The decimal expansion of $\frac{87}{160}$ is: 1

(a) an integer **(b)** non-terminating non-repeating

(c) terminating **(d)** non-terminating repeating

10. If $\sqrt{13}$ is irrational, then $\sqrt{13} \times \sqrt{13}$ is: 1

(a) irrational **(b)** rational

(c) not defined **(d)** non-terminating non-repeating

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Without actual division, find how many decimal places $\frac{31}{625}$ has. 2

12. Find the largest number which divides 1251 and 9377 leaving remainders 1 and 2 respectively. 2

13. Find the smallest number which leaves remainder 7 when divided by 8, 18 and 24. 2

14. Without actual division, find how many decimal places $\frac{9}{500}$ has. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The HCF of two numbers is 6 and their LCM is 396. If one number is 66, find the other. 3

16. Three bells toll at intervals of 10, 16 and 24 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

17. Show that $(9 + 3\sqrt{5})(9 - 3\sqrt{5})$ is rational but $9 + 3\sqrt{5}$ is irrational. 3

OR

The HCF of two numbers is 14 and their LCM is 420. If one number is 70, find the other. 3

18. Three bells toll at intervals of 14, 18 and 21 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

OR

Show that $(4 + 1\sqrt{11})(4 - 1\sqrt{11})$ is rational but $4 + 1\sqrt{11}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two tankers contain 468 litres and 624 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 84 m by 56 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

20. Two tankers contain 720 litres and 960 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

A rectangular courtyard is 90 m by 54 m. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

END OF QUESTION PAPER — Total: 40 Marks

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